

HajjSim Agent Studio

A Digital Twin and Multi-Agent Simulation for Hajj Crowd Safety

Modeling pilgrim movement, health stress, crowd pressure, and ritual progress



SAFETY



MULTI-AGENT
SIMULATION



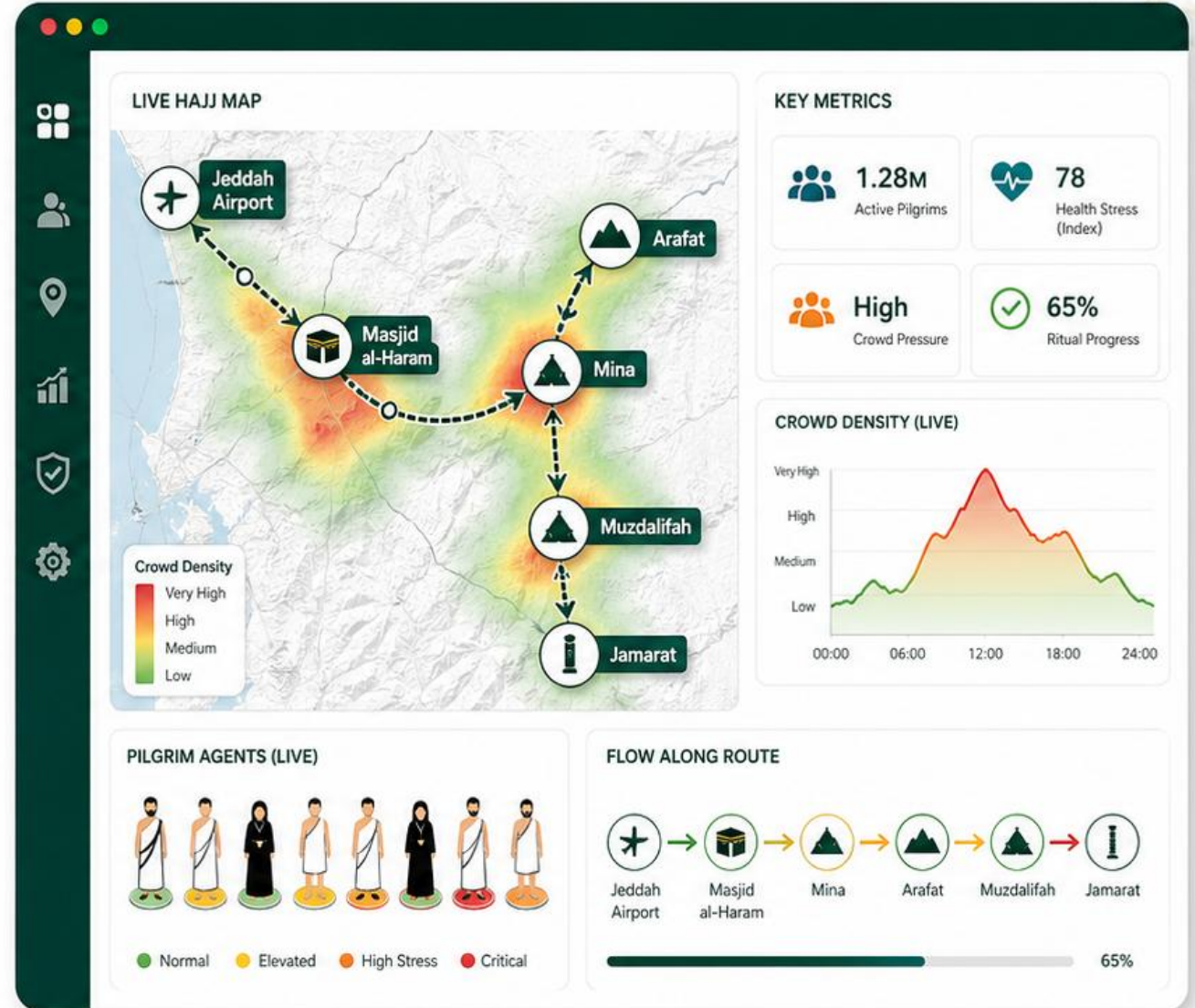
DIGITAL TWIN



ANALYTICS

Prototype Project Presentation

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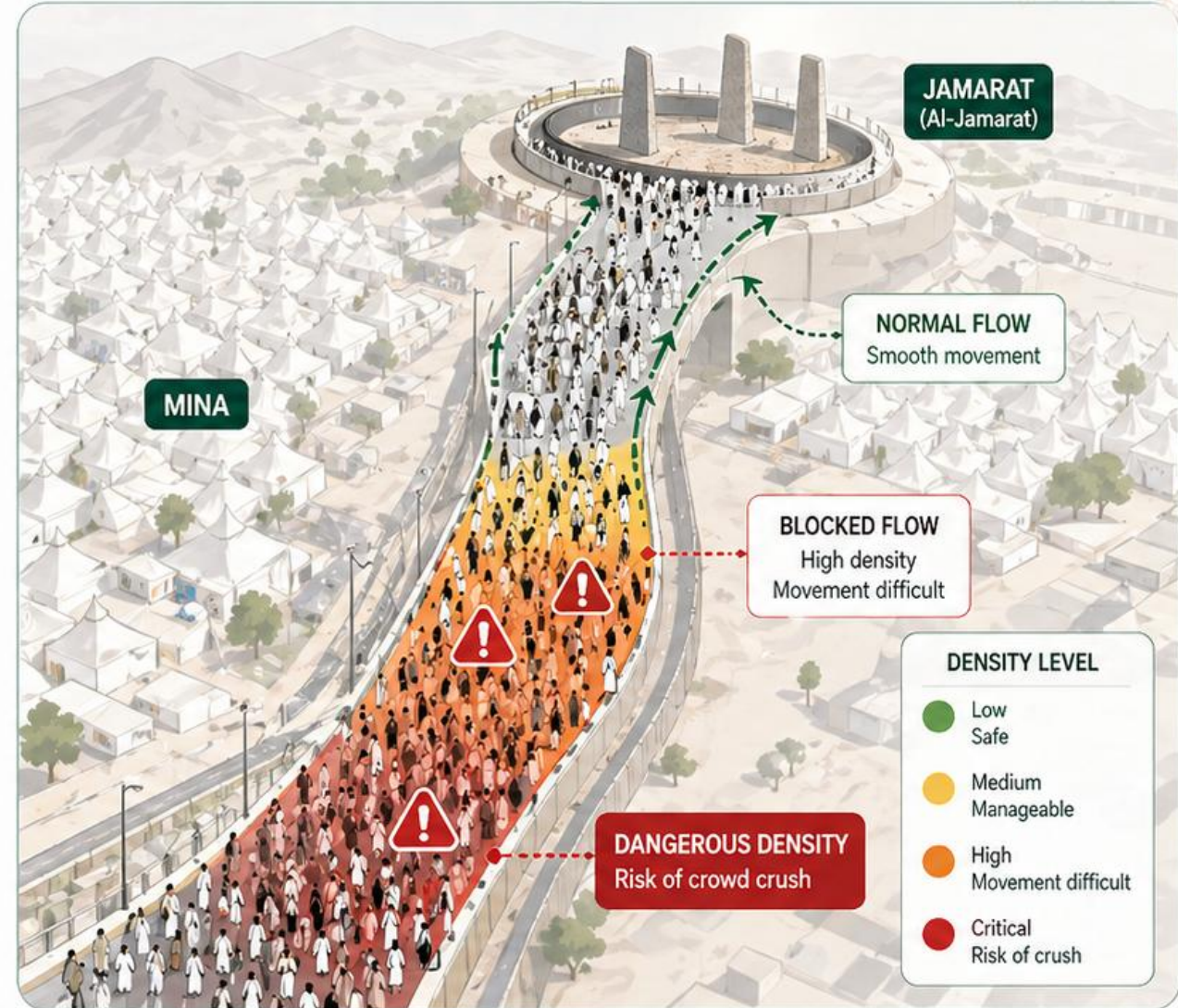
Real-World Motivation

The Mina Crowd Crush in 2015

-  A major crowd crush happened in Mina during Hajj
-  Very high density made movement difficult
-  Some pilgrims were trapped in the crowd
-  Many people died and many were injured
-  The tragedy showed the need for better crowd prediction and monitoring



When crowd density becomes too high, people may not be able to move or breathe.



A safer Hajj needs better visibility before crowd pressure becomes critical.

Why This Project Matters

- ✓ Millions of pilgrims move in limited spaces
- ✓ Many rituals happen at similar times
- ✓ Heat, fatigue, and crowd pressure increase risk
- ✓ Small problems can become dangerous very fast
- ✓ Operators need tools that warn early



High Density

Too many people in small spaces



Heat Stress

High temperature and fatigue raise health risks



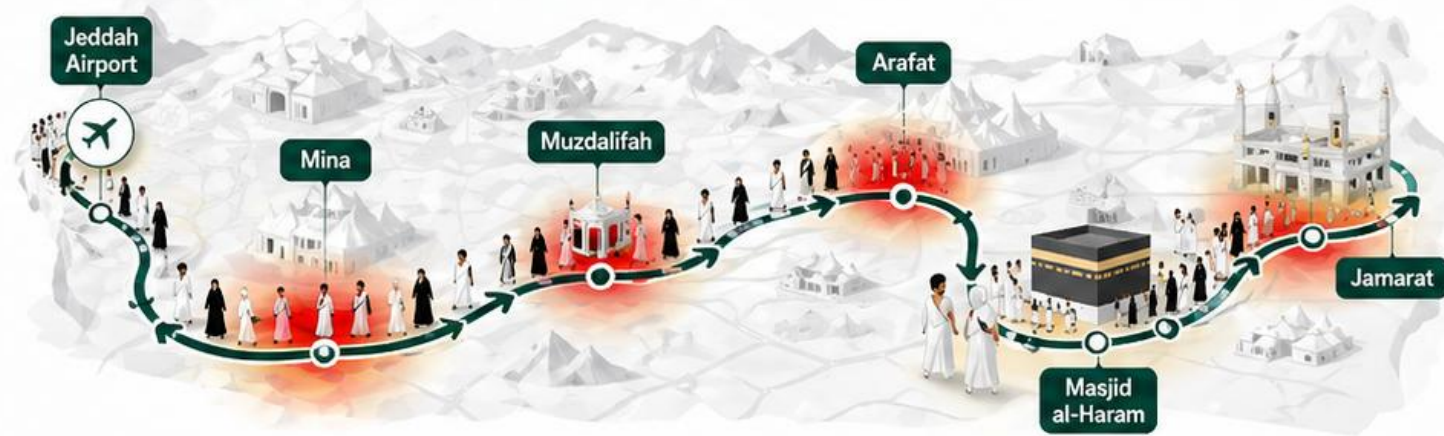
Route Congestion

Bottlenecks slow movement and create pressure



Emergency Risk

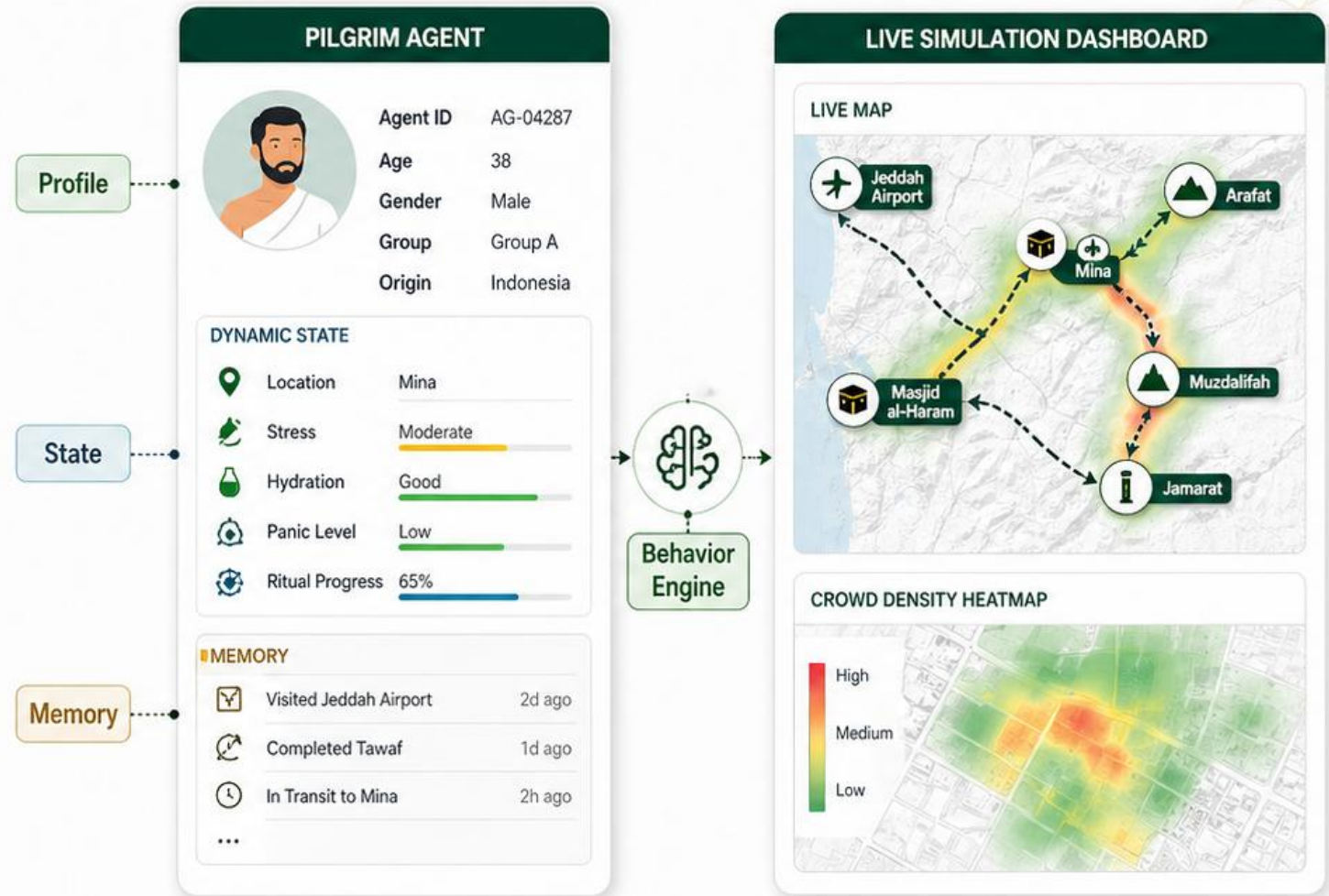
Small incidents can escalate quickly in crowded areas



Hajj needs proactive planning, not only reaction after problems happen.

What Is HajjSim Agent Studio?

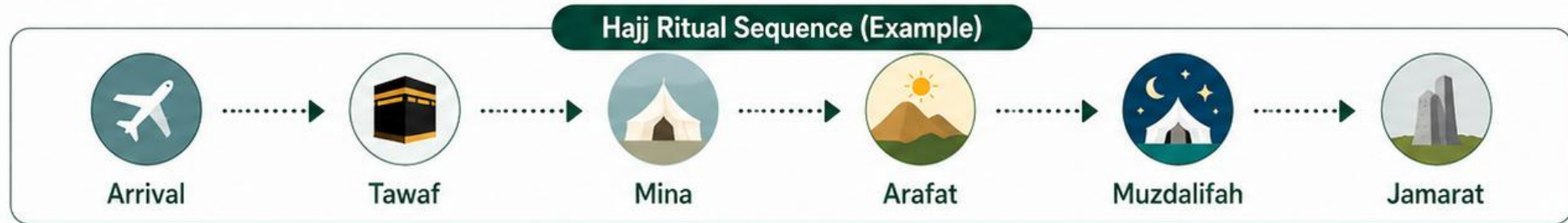
- A digital twin of the Hajj journey
- Each pilgrim is modeled as an intelligent software agent
- The system tracks movement, stress, hydration, panic, and ritual progress
- Users can test normal situations and emergency scenarios
- The goal is to understand behavior before problems happen



From data to insight for safer Hajj operations.

How the Simulation Works

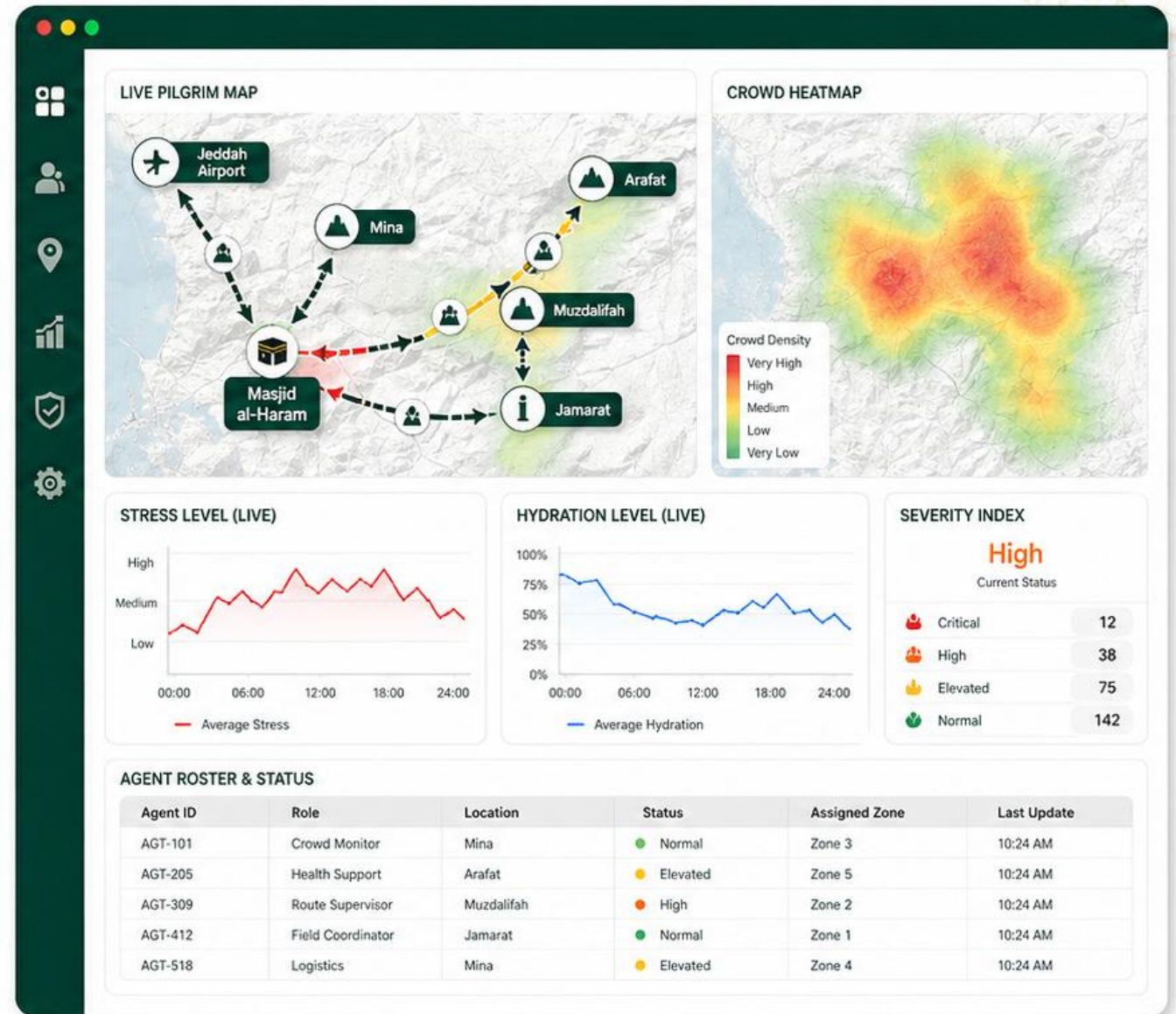
- Each simulation tick updates the agent state
- Agents follow the real Hajj ritual sequence
- Examples: arrival, Tawaf, Mina, Arafat, Muzdalifah, Jamarat
- This creates a clear spatial and time-based model



The model turns pilgrim behavior into a testable simulation.

What the Dashboard Shows

- Live pilgrim locations on a map
- Heatmap of crowd pressure
- Stress and hydration metrics
- Route overlays and status colors
- Operational summary and agent roster



Users can run the simulation step by step or in continuous mode.

Why It Helps Hajj Operations



Detect Bottlenecks

Find crowded routes before they become dangerous



Protect Vulnerable Pilgrims

Spot groups with high stress or health risk



Test Emergency Plans

Simulate panic, heat, or medical incidents safely



Support Better Decisions

Give planners and operators useful live insight



Improve Safety

Reduce risk and improve movement flow



The system helps teams act earlier and smarter.

Future Plan

How We Will Improve the System



Leader Agents

Guide groups, keep members together, and share instructions



Transportation Agents

Model buses, shuttles, delays, and movement between sites



Housing Agents

Manage camps, hotels, room flow, and accommodation pressure



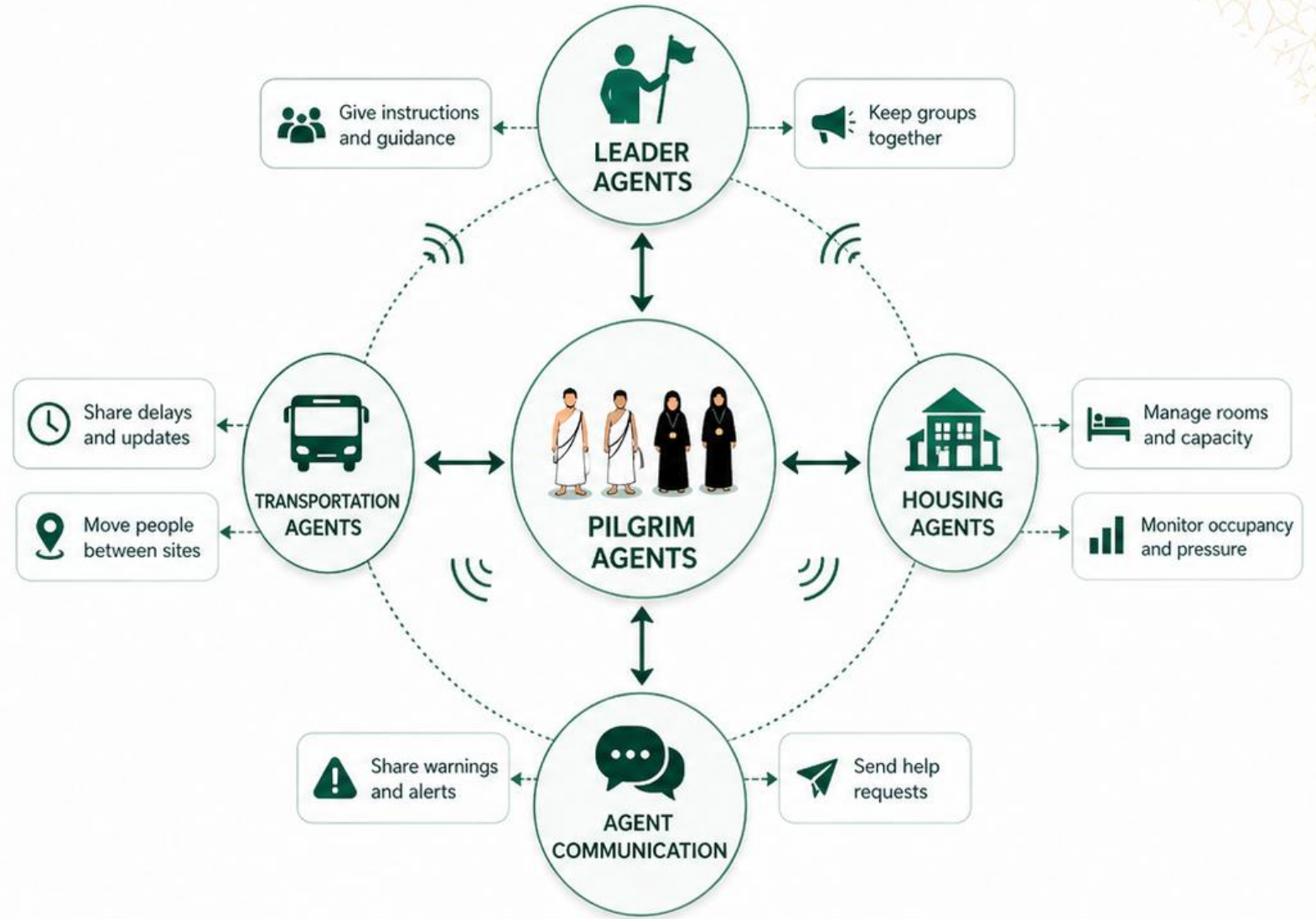
Agent Communication

Let agents share warnings, route updates, and help requests



Next Stage

Use more agents, larger populations, and smarter prediction



The long-term goal is a smarter and more realistic Hajj decision-support platform.

References

1. Reuters. Death toll in Saudi haj disaster at least 2,070, Reuters tally. 29 Oct 2015.
2. Associated Press. AP count: Over 2,400 killed in Saudi hajj stampede, crush. 9 Dec 2015.
3. Salamati P, et al. Hajj Stampede in Mina, 2015: Need for Intervention. 2016.
4. Alaska YA, et al. The impact of crowd control measures on the occurrence of stampedes. 2017.
5. Project source: HajjSim Agent Studio prototype description provided by the project team.

These references support the historical motivation slide. The project description and technical content come from the provided HajjSim brief.